

FEA of Thin-Walled Circular Tube

1. Problem Description

1.1 Objective

This report aims to carry out a Finite Element Analysis (FEA) study on a thin-walled cantilever circular beam. Responses of the beam to different modes of failure as defined in section 1.3 are investigated and results are summarised. Based on findings, conclusions are drawn.

1.2 Model Setup

The structure analysed is a thin-walled steel cylindrical pipe modelled as a shell. Overall length $L = 5.0$ m, outer diameter $d_{outer} = 500$ mm, wall thickness $t = 8$ mm (inner diameter $d_{inner} = 484$ mm). A concentrated transverse load/displacement is applied at a point $a = 4.5$ m from the fixed end (i.e., 0.5 m from the free end). The pipe was modelled using shell elements with a thickness of 8 mm. Refer to Figure 7.

A circular load patch of diameter 50 mm was created on the shell surface. Remote loading was applied using a distributed coupling to avoid artificial stress singularities. A local mesh control of 10 mm was applied on this patch.

The material properties used are shown in Figure 3.

1.3 Definition of failure modes

Three widely used failure limits were investigated. A wider range of possible outcomes have been explored for accurate comparison.

- Yield Initiation – Occurs when maximum von Mises Stress reaches yield stress. The failure criteria in effect uses irreversible deformation as a marker for initiation of failure. (Christensen, 2001)
- Elastic Buckling – An elastic column may buckle with lateral deflection before yielding, if the compressive force is large enough causing it to fail at a critical load. These critical loads or eigenvalues can be used to predict deformation. (Bertram & Gluge, 2015)
- Brazier Ovalisations – A perfectly circular tube, under bending, tends to ovalise or deform into an ellipse. Ovalisation leads to a reduction in the second moment of area which can directly affect behaviour under load. (Knaster, et al., 2003)

2. Model verification

2.1 Theoretical validation

Simple bending theory equations can be used to deduce starting parameters.

Second Moment of Area:

$$I = \frac{\pi}{64} (d_{outer}^4 - d_{inner}^4) \quad (1)$$

Where I is the Second moment of Area in m^4 , d_{outer} is the outer diameter, and d_{inner} is the inner diameter of the cantilever beam in m.

$$I = \frac{\pi}{64} (0.5^4 - 0.484^4) = 3.7425 \times 10^{-4} m^4 \quad (2)$$

Bending Moment at fixed end:

$$M = FL \quad (3)$$

Where M is the Bending Moment in Nm, F is the Force or load in N, and L is the perpendicular distance between the force and the fixed end in m.

Bending stress:

$$\sigma = M \frac{d_{outer}}{2I} \quad (4)$$

From (2):

$$\sigma = FL \frac{d_{outer}}{2I} \quad (5)$$

Hence, stress per N:

$$\sigma = F \times 4.5 \times \frac{0.5}{2 \times 3.7425 \times 10^{-4}} = 3006 \text{ Pa per N} \quad (6)$$

Where σ is the stress in Pa

Therefore, for an arbitrary force value of 50 kN:

$$\sigma = 50000 \times 4.5 \times \frac{0.5}{2 \times 3.7425 \times 10^{-4}} = 150.3 \text{ MPa} \quad (7)$$

For comparison with FEA, simple Euler–Bernoulli beam theory was used (small deflection, linear-elastic, uniform section).

Tip Deflection estimate for 50 kN:

$$\delta = \frac{FL^3}{3EI} = 20.29 \times 10^{-3} \text{ m} = 20.29 \text{ mm} \quad (8)$$

Where δ is the maximum deflection of the tip in m

With $E = 200 \text{ GPa}$, $\delta(a) \approx 20.3 \text{ mm}$. (Displacement was extracted at the load application point, not at the free end.)

A 50 kN Linear Static study was thus conducted and the findings are shown in Figure 8.

Percentage difference between theoretical and simulated stress values:

$$\% = \frac{151.3 - 150.3}{150.3} \times 100 = 0.66\% \quad (9)$$

Percentage difference between theoretical and simulated displacement:

$$\% = \frac{20.29 - 19.82}{19.82} \times 100 = 2.37\% \quad (10)$$

This clearly validates the theoretical predictions and therefore can be used for further analysis.

2.2 Mesh setup

Mesh settings were chosen based on result accuracy and sensible computational time using the static linear study with 80 kN load as a baseline. Based on Table 1, a global mesh element size of 60 mm was chosen. A localised mesh size between 5-10 mm was used at the load patch for controlled output. Curvature-Based mesh parameters were chosen due to consistency between studies. Coarser meshes were not considered due to adequate availability of computing resources for fine mesh simulations.

Notes on convergence: Tip displacement changes from 34.09 mm (80 mm mesh) to 36.01 mm (60 mm mesh) to 36.12 mm (40 mm mesh), indicating displacement is close to converged by 60–40 mm. Maximum von Mises stress increases from 223.5 MPa to 245.7 MPa to 255.4 MPa; peak stress is not fully converged because it is sensitive to local refinement around the load application region.

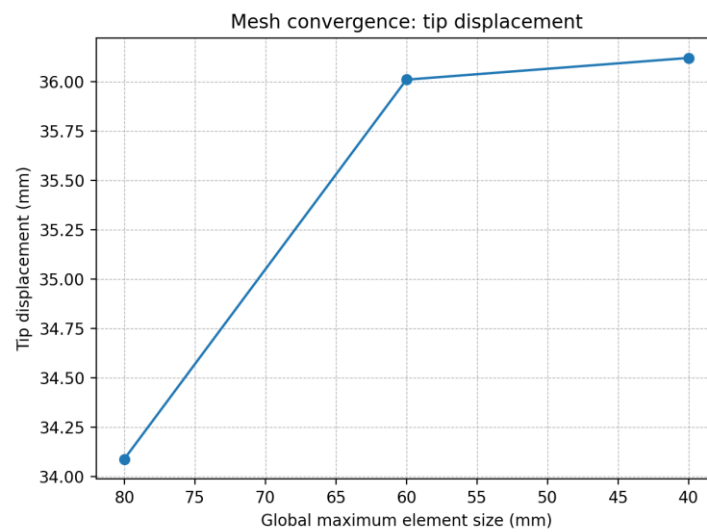


Figure 1: Mesh convergence of tip displacement with global maximum element size.

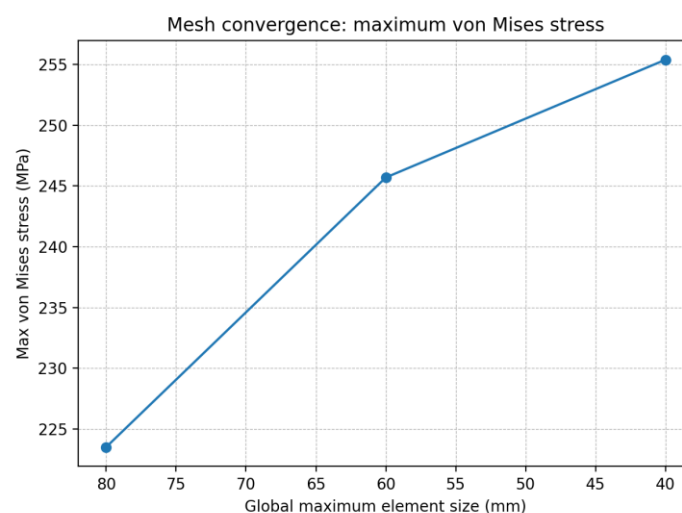


Figure 2: Mesh convergence of maximum von Mises stress with global maximum element size.

Table 1: Mesh Study.

Mesh type	Max element size (mm)	No. elements	Tip deflection (mm)	Max von Mises (MPa)
Coarse	80	8564	34.09	223.5
Medium	60	17239	36.01	245.7
Fine	40	33654	36.12	255.4

As results were validated, yield initiation load can be estimated.

From (4), using given yield stress of 235 MPa:

$$F = 235000000 \times \frac{2 \times 3.7425 \times 10^{-4}}{4.5 \times 0.5} = 78.18 \times 10^3 N = 78.2 \text{ kN} \quad (10)$$

This value can be used to simulate the first mode of failure addressed, namely yield initiation. Further assumptions can be made to simulate the other two failure types. These are discussed in detail in the next section.

3. Results and Discussion

3.1 Yield Initiation

A linear static yield study was conducted at a remote load of 80 kN. The load was localized to a $\varnothing$ 50 mm patch. This produced a stress singularity at the point of contact. The maximum stress of 559 MPa shows that the load is effectively applied through a tiny region. This can be considered abnormal as the minimum stress is much larger than the maximum yield stress of 235 MPa. Refer to Figure 11.

A new study with the exact same parameters except loading was conducted. This time the force was loaded as a distributed connection across the whole cylindrical face. The corrected results align closely with theoretical predictions, confirming that yield initiation is a credible governing failure mode for this structure. Refer to Figure 12.

Linear static response shows a proportional relationship between force and displacement, consistent with elastic beam theory. Nonlinear static response begins to deviate slightly from linearity as stresses approach yield, indicating stiffness reduction due to material nonlinearity.

3.2 Buckling

A buckling study was set up using the same loading and mesh parameters. 10 modes were considered and load factor of each mode shape was examined. Mode 4 produced the smallest magnitude load factor of 64.12. This load factor signifies a very high critical load of 3 MN (64 times the load applied). Compared to the first yield load of 80 kN, this buckling load is way higher. Hence, eigenvalue buckling does not govern failure in this case. Eigenvalue buckling predicts ideal elastic instability and therefore overestimates real buckling. Refer to Figure 14.

Table 2: Buckling loading Factor

Mode	Loading Factor
1	-97.37
2	-92.24
3	-81.59
4	64.12
5	68.61
6	89.44
7	95.46
8	98.72
9	99.91
10	101.75

3.3 Ovalisations

To produce Brazier effect/ovalisations, a nonlinear static study was conducted as it was found to be unachievable with a Linear Static study. Loading and mesh setup were kept constant for accurate comparison against the validated 50 kN Linear Static Study. Remote Displacement was also applied but this showed negligible difference when compared to Remote Load and hence was not considered for further analysis. This produced a maximum stress higher than the yield stress at 50 kN. While some cross-sectional distortion is visible in deformed plots, clear quantitative ovalisation (change in major/minor diameters) was difficult to extract reliably.

This is due to dominance of local bending stresses near the load patch and material yielding occurring before significant ovalisation develops. As such, Brazier ovalisation is not identified as the governing failure mode for this configuration. Refer to Figure 15.

4. Conclusion

Multiple studies were conducted to analyse failure behaviour. Agreement between theoretical predictions and baseline static study is strong, which supports correctly applied boundary conditions and parameters. For thin-walled shells, buckling may be used to identify mode shapes rather than definitive failure load. Non-Linear analysis did not produce desirable ovalisations. Based on these results, yield initiation is considered the primary failure mode as this produced the most reliable outputs.

Overall, the modelling approach, assumptions, and results are consistent with thin-shell structural predictions. Limitations related to stress singularities, nonlinear convergence, and ovalisation measurement are clearly identified.

5. References

- Bertram, A. & Gluge, R., 2015. Buckling. In: *Solid Mechanics*. s.l.:Springer, pp. 222-228.
- Christensen, M. R., 2001. *Defining Yield Stress and Failure Stress*. [Online]
Available at:
[https://www.failurecriteria.com/Media/Defining_Yield_Stress_and_Failure_Stress_\(Strength\).pdf](https://www.failurecriteria.com/Media/Defining_Yield_Stress_and_Failure_Stress_(Strength).pdf)
- Knaster, J., Tortora, G. & Veness, R., 2003. *THE BRAZIER EFFECT AND ITS INFLUENCE IN THE DESIGN OF BEAMPIPES FOR PARTICLE COLLIDERS*, Geneva: CERN.

Timoshenko, S.P. and Gere, J.M. (1961) Theory of Elastic Stability. 2nd edn. New York: McGraw-Hill.

6. Appendix

6.1 Model Setup

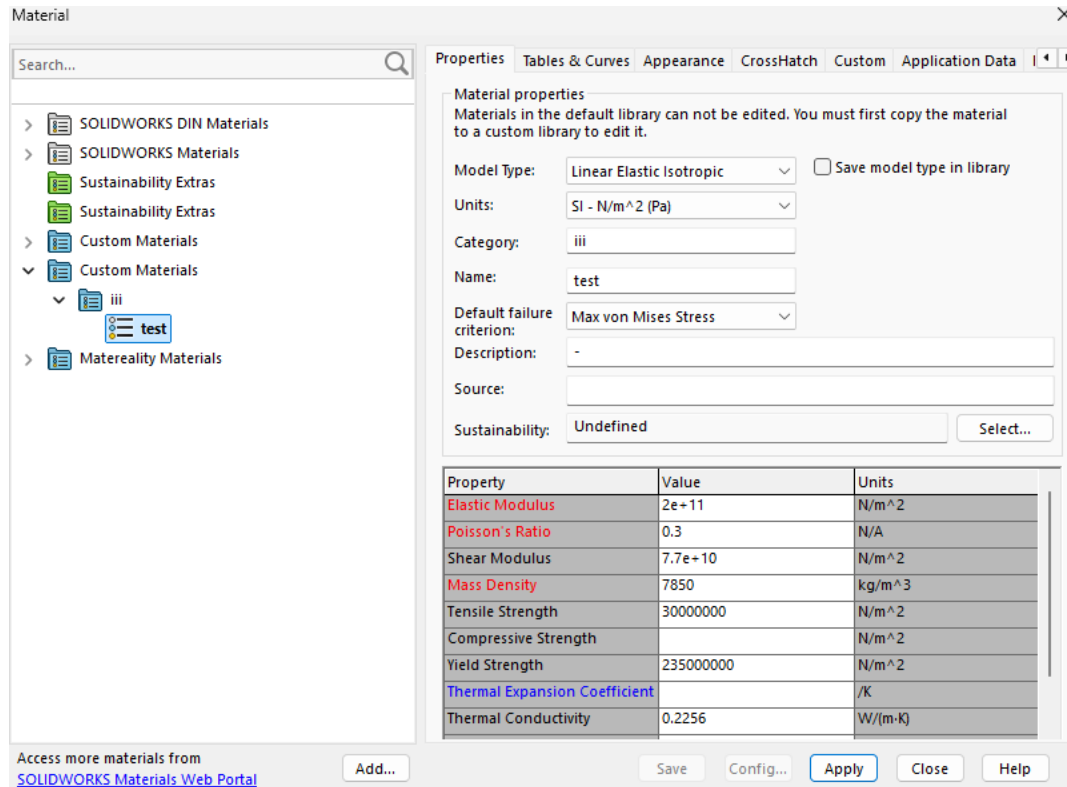


Figure 3: Custom material properties.

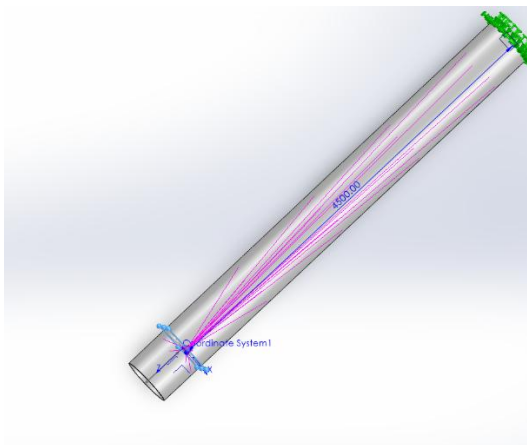


Figure 4: Pipe geometry and load location.

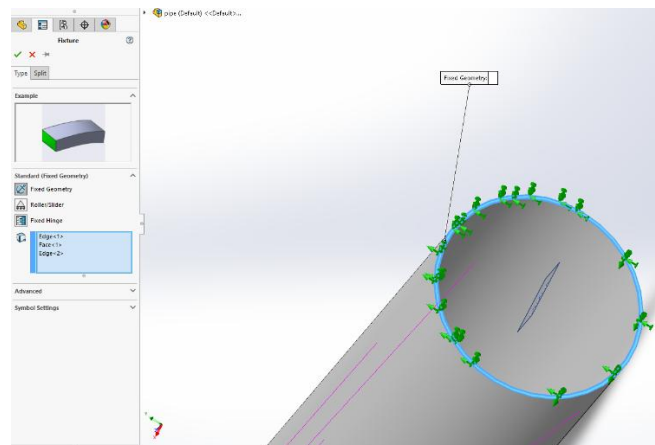


Figure 5: Fixed Geometry applied to fixed end.

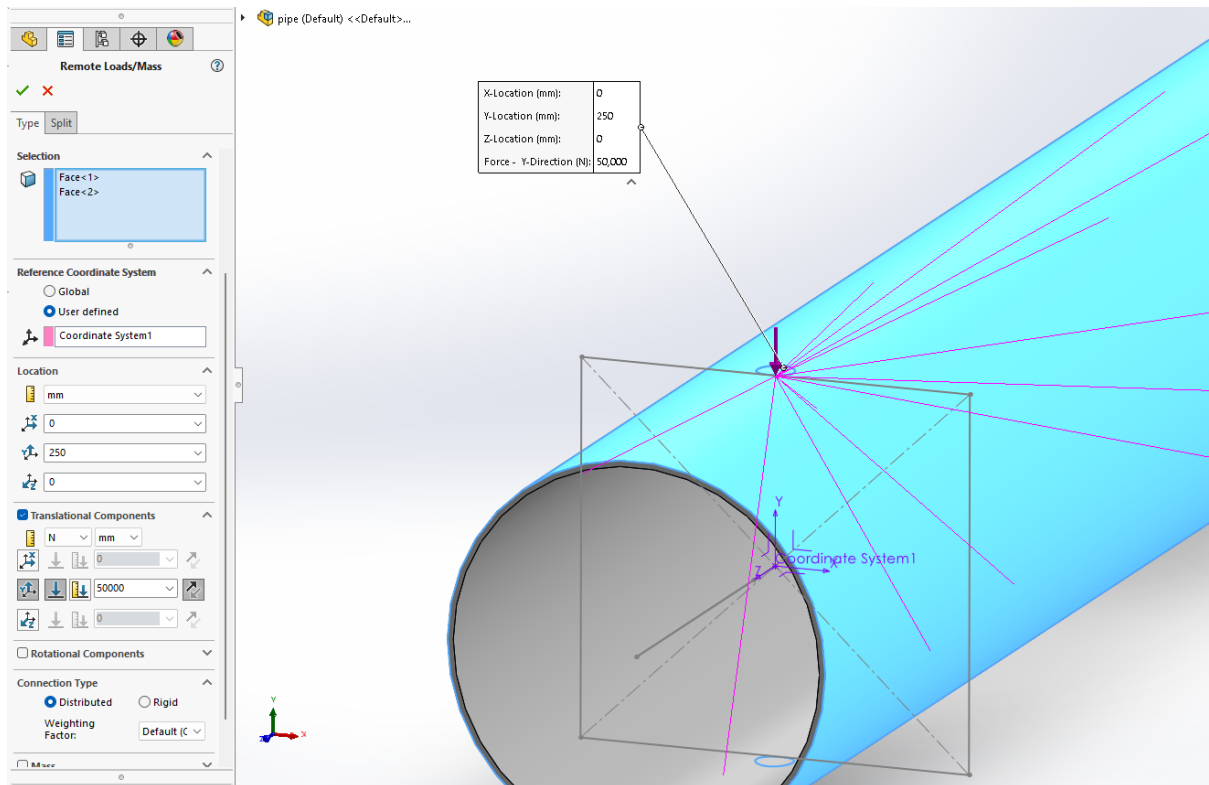


Figure 6: Remote Load setup (distributed connection) at the load point.

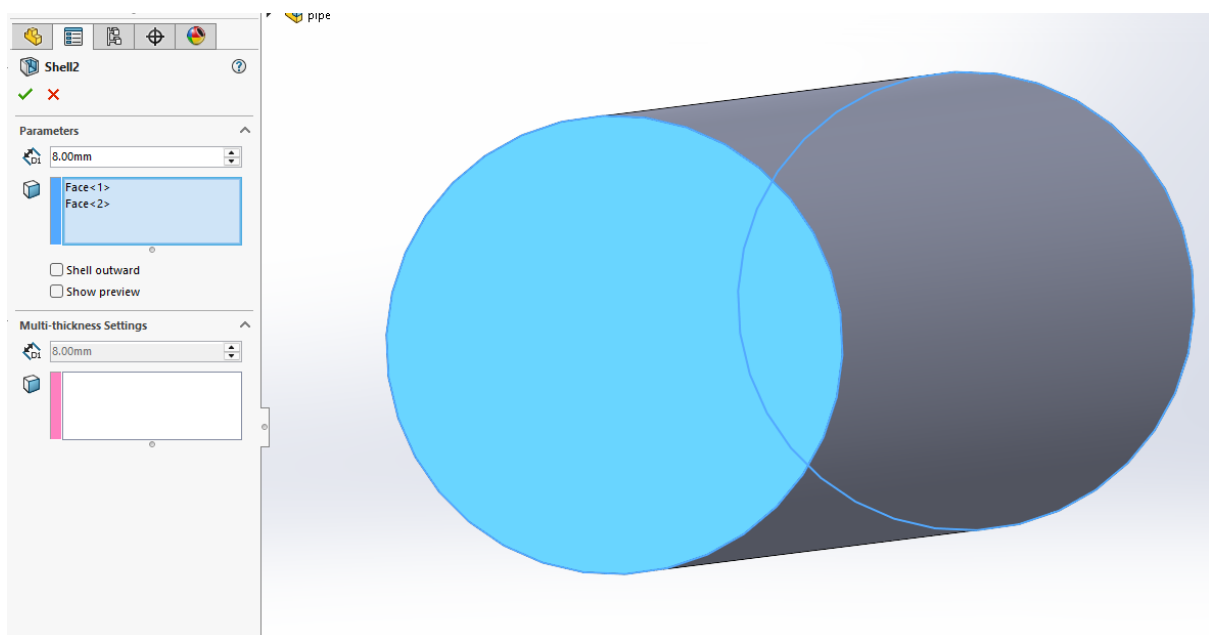


Figure 7: Shell feature used to model tube.

6.2 Reports and Additional information

Study Results

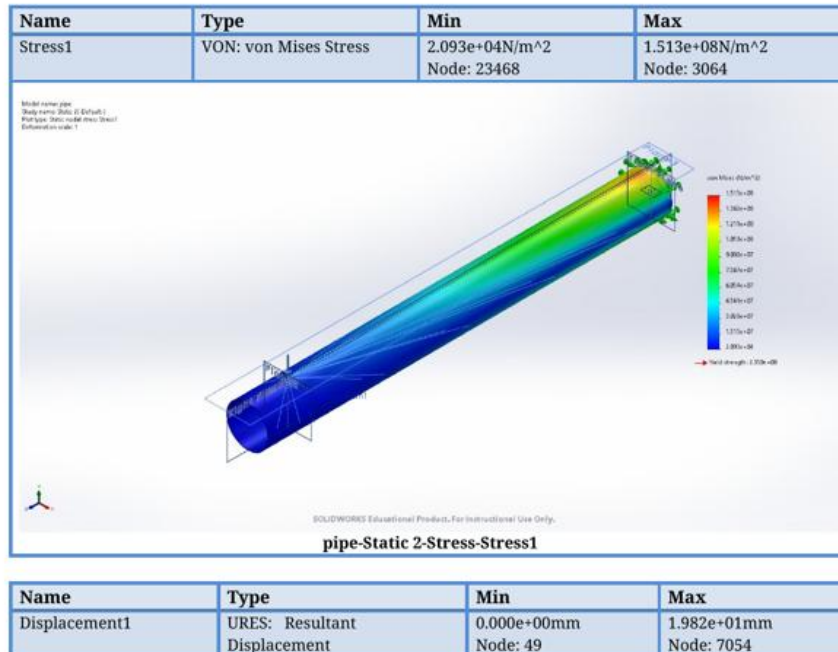


Figure 8: FEA Results 50 kN Linear.

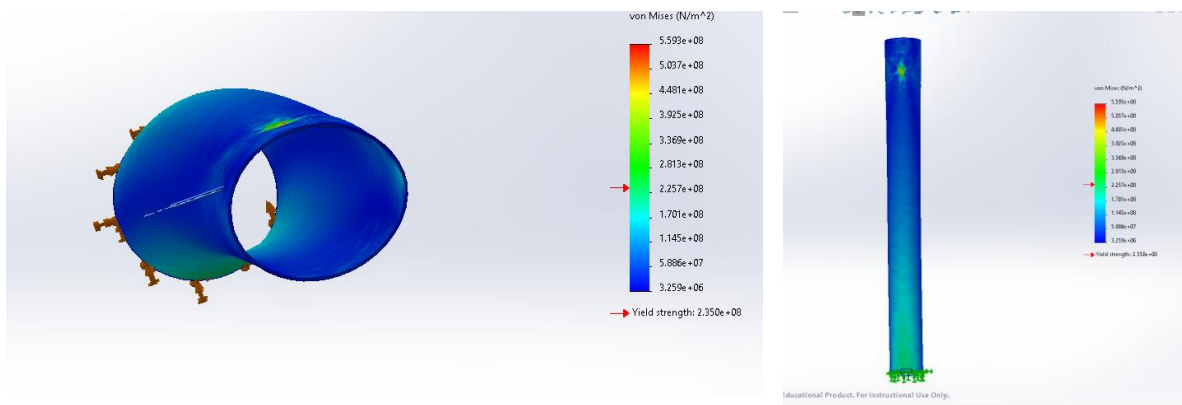


Figure 9: (a) 80 kN Linear FEA Isometric View; (b) 80 kN Linear FEA Top View

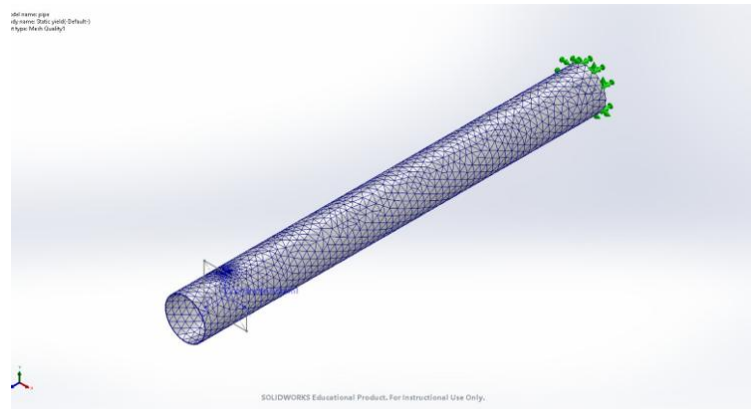


Figure 10: Mesh Plot.

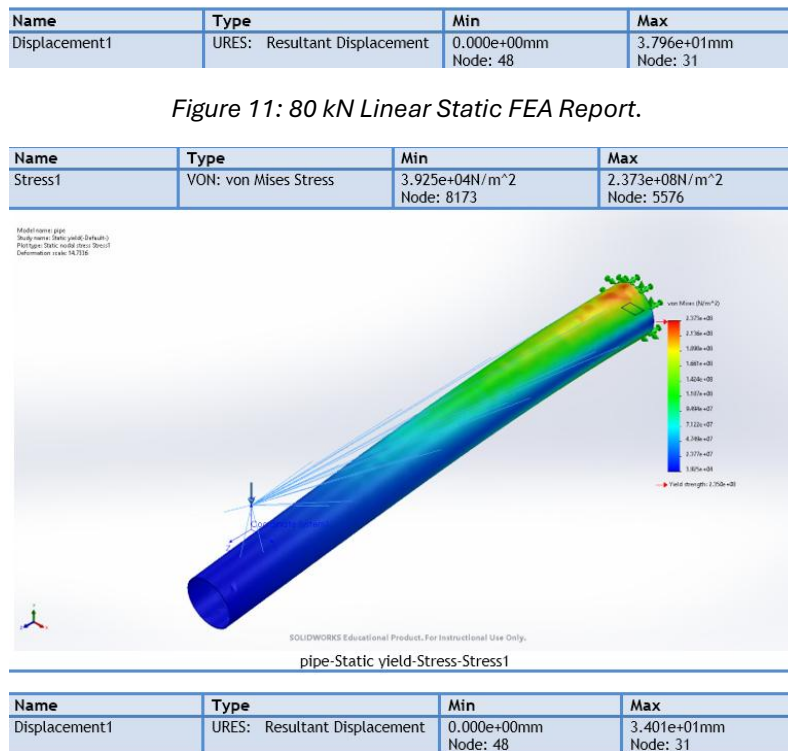
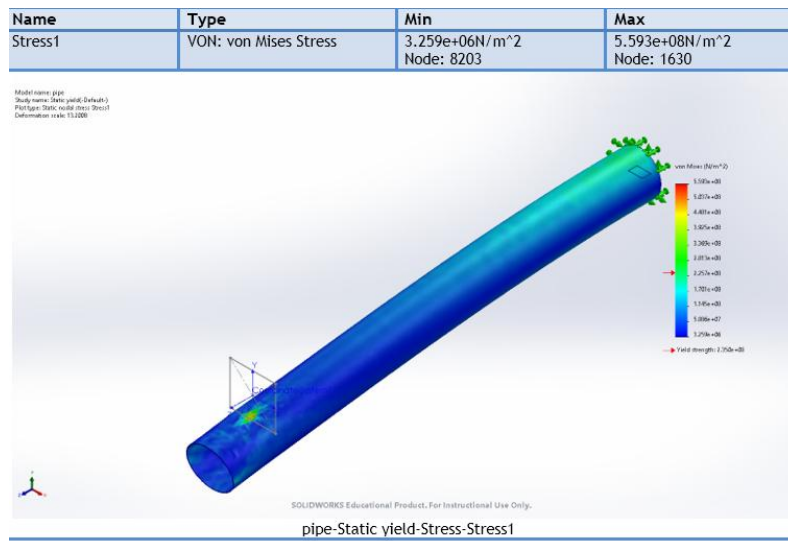


Figure 12: UPDATED 80 kN Linear Static FEA Report.

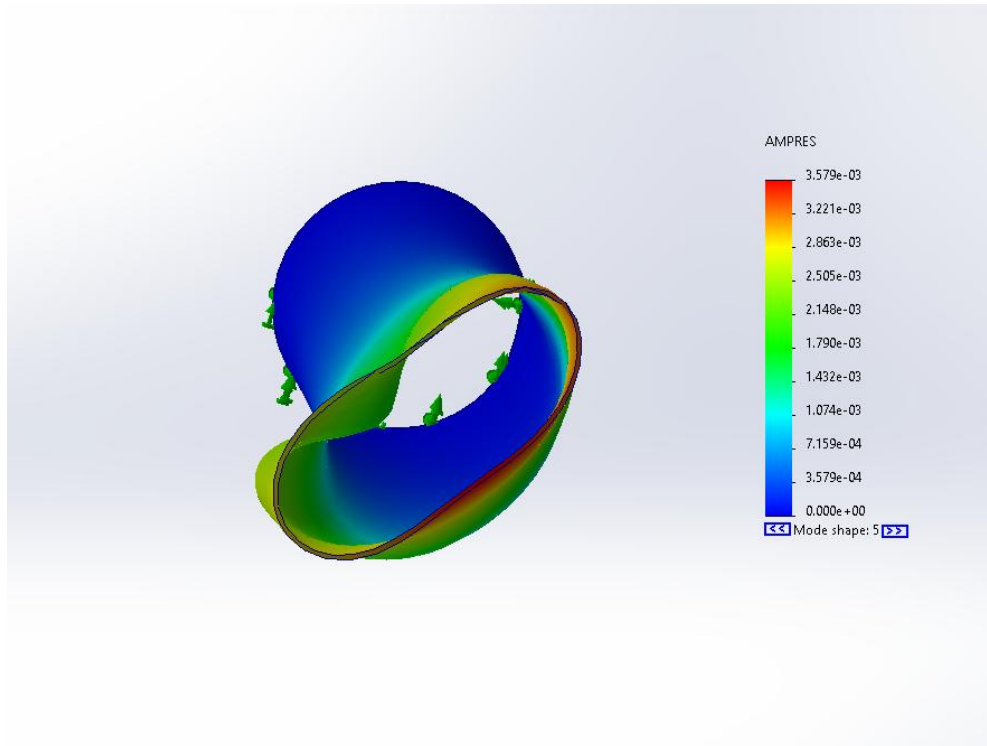


Figure 13: Buckling FEA stress distribution.

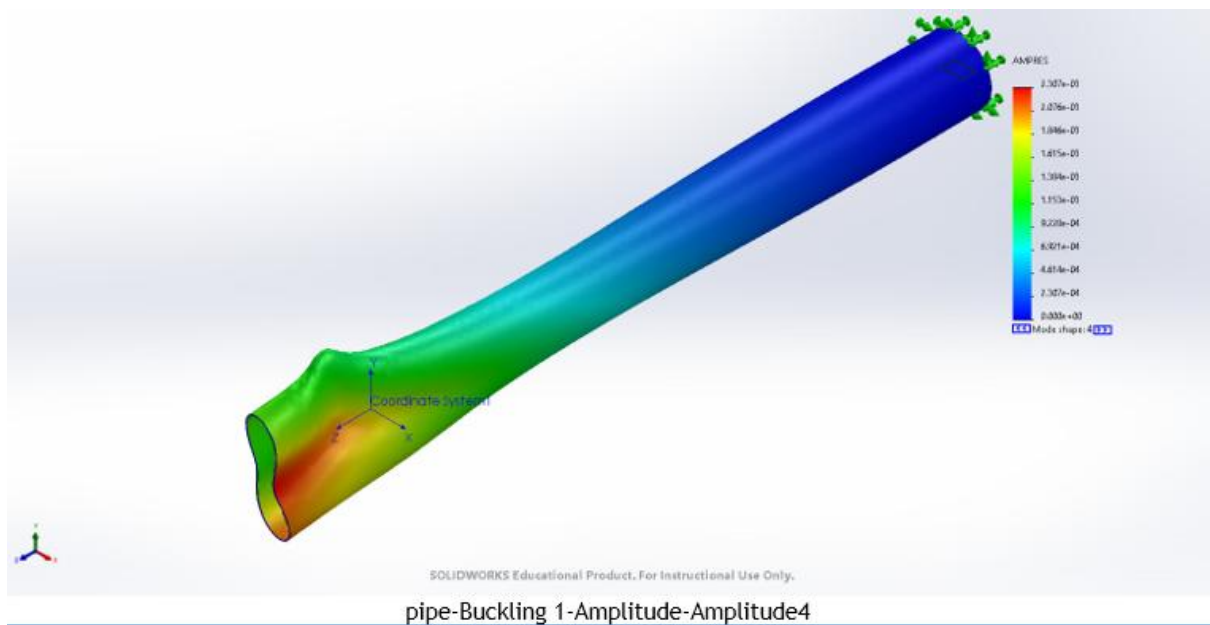


Figure 14: Buckling FEA stress distribution Isometric View.

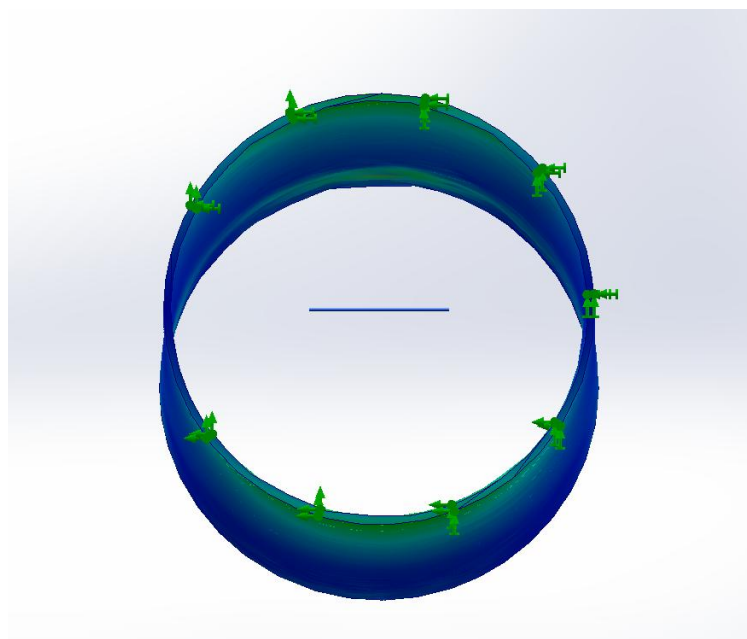
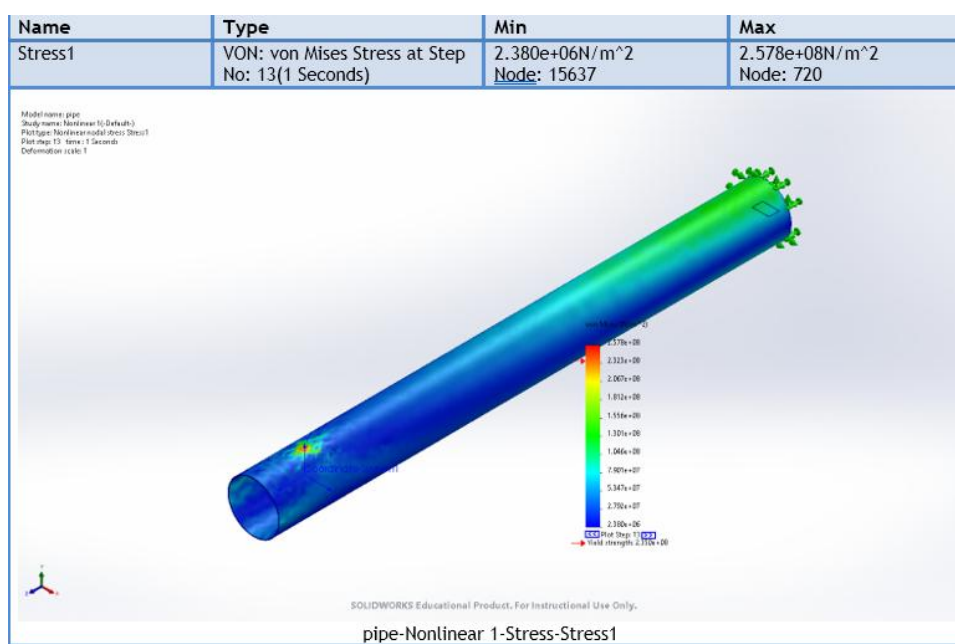


Figure 15: Nonlinear Static FEA Ovalisations.



Name	Type	Min	Max
Displacement1	URES: Resultant Displacement at Step No: 13(1 Seconds)	0.000e+00mm Node: 48	2.394e+01mm Node: 31

Figure 16: Nonlinear Static FEA Report.